

VERHNYAYA TOYMA, Russian Federation

WMO: 227780

Lat: 62.2412N

Lon: 45.0131E

Elev: 58

StdP: 100.63

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.3	-29.7	-36.1	0.1	-33.3	-32.5	0.2	-29.6	8.7	-5.5	8.0	-7.0	1.3	30	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	27.6	18.6	25.6	17.6	23.8	16.6	19.9	25.3	18.8	23.8	17.7	22.2	3.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.0	22.4	16.9	12.1	21.1	15.9	11.3	20.0	57.2	25.2	53.7	23.9	50.2	22.2	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.9	6.2	-36.0	29.9	3.1	1.8	-38.2	31.2	-40.0	32.2	-41.8	33.2	-44.0	34.5		
(5)			-36.1	21.5	2.9	1.5	-38.2	22.6	-39.9	23.5	-41.5	24.4	-43.6	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-13.2	-10.6	-5.8	1.5	8.4	14.3	16.8	14.2	8.9	1.9	-5.0	-9.6		
	DBStd	11.77	8.56	8.71	5.98	5.09	5.47	4.33	3.61	3.75	3.60	4.07	6.33	8.19		
	HDD10.0	3538	721	577	491	259	102	10	1	7	62	250	451	608		
	HDD18.3	6058	979	811	749	505	312	135	74	137	282	508	701	867		
	CDD10.0	570	0	0	0	4	51	138	211	136	29	1	0	0		
	CDD18.3	49	0	0	0	0	2	13	25	8	0	0	0	0		
	CDH23.3	476	0	0	0	1	29	153	223	69	1	0	0	0		
	CDH26.7	89	0	0	0	0	3	36	41	9	0	0	0	0		
(14) Wind	WSAvg	3.0	3.0	3.3	3.2	3.0	3.3	3.0	2.8	2.7	2.7	3.2	3.1	3.0		
(15) Precipitation	PrecAvg	586	37	32	25	36	46	57	61	67	55	62	51	47		
	PrecMax	666	54	59	43	75	75	117	124	119	111	140	128	72		
	PrecMin	503	26	10	4	9	18	16	21	27	23	9	21	23		
	PrecStd	45	8	13	10	21	15	27	33	27	24	31	25	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	2.6	6.6	19.9	26.7	30.5	30.4	28.1	21.5	11.7	5.9	2.9		
		MCWB	0.5	1.5	2.3	11.7	16.4	20.4	19.6	19.5	16.1	8.9	5.5	2.2		
	2%	DB	0.3	1.6	4.1	14.5	23.4	27.1	27.9	25.2	19.0	9.8	3.7	1.5		
		MCWB	-0.1	0.8	1.3	8.3	14.2	17.9	18.9	18.0	14.4	8.4	3.2	0.9		
	5%	DB	-0.8	0.7	2.7	10.6	20.9	24.8	26.1	23.0	16.9	8.8	2.7	0.8		
		MCWB	-1.1	0.0	0.6	5.8	13.2	16.8	18.2	17.1	13.3	7.6	2.3	0.4		
	10%	DB	-2.6	-0.5	1.7	8.4	18.2	22.5	24.2	21.0	14.9	7.5	1.6	0.1		
		MCWB	-3.0	-1.0	0.2	4.4	11.8	15.5	17.0	16.1	12.1	6.4	1.0	-0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	2.0	3.0	11.6	18.2	21.6	21.7	20.4	16.9	9.7	5.6	2.2		
		MCDB	1.0	2.5	5.4	19.1	23.7	28.2	27.5	25.2	19.5	10.7	5.8	2.6		
	2%	WB	0.0	0.8	1.9	8.6	15.9	19.6	20.1	19.1	15.3	8.8	3.3	1.1		
		MCDB	0.4	1.4	3.6	12.6	20.7	25.1	25.8	24.1	18.2	9.7	3.6	1.3		
	5%	WB	-1.2	0.1	1.1	6.4	14.1	18.1	19.1	17.9	13.7	7.7	2.5	0.5		
		MCDB	-0.9	0.7	2.3	9.8	18.9	22.8	24.0	21.7	16.1	8.7	2.8	0.8		
	10%	WB	-2.9	-0.9	0.4	5.0	12.4	16.6	18.0	16.7	12.4	6.5	1.2	-0.2		
		MCDB	-2.6	-0.4	1.6	7.8	18.0	21.2	22.9	20.1	14.5	7.5	1.5	0.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.6	9.0	8.8	10.2	10.2	10.0	9.1	7.8	4.6	4.3	5.4		
		MCDBR	5.0	4.5	7.9	12.3	14.1	14.1	13.0	12.2	11.5	6.0	3.4	4.1		
	5% WB	MCWBR	4.9	4.3	5.6	7.3	7.1	6.6	5.5	6.1	7.2	5.1	3.2	4.1		
		MCWBR	4.9	4.8	5.9	10.6	11.8	11.9	11.2	10.8	9.7	5.9	3.4	4.2		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.295	0.326	0.360	0.354	0.354	0.391	0.371	0.337	0.316	0.262	0.221		
	taud		1.989	2.058	2.081	2.168	2.439	2.474	2.377	2.408	2.490	2.369	2.083	1.943		
	Ebn at Noon		397	630	760	818	866	872	822	808	773	634	380	236		
	Edh at Noon		37	74	104	118	99	97	105	94	72	55	34	21		
(44) All-Sky Solar Radiation	RadAvg		0.24	0.86	2.17	3.63	4.86	5.19	5.21	3.61	2.00	0.75	0.22	0.11		
	RadStd		0.04	0.21	0.38	0.44	0.30	0.57	0.42	0.37	0.22	0.11	0.05	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page